

Extracting Data using SQL.

Recap:

RDBMS.

DB

P.K., F.K.

Schema

Data types

E-R diagram

CRUD

SQL VS NOSQL.

Agenda

Relationships

ER-diagram

Queries.

Select

From

Order by

⋮

Relationships.

4. 1 - 1

1 - many

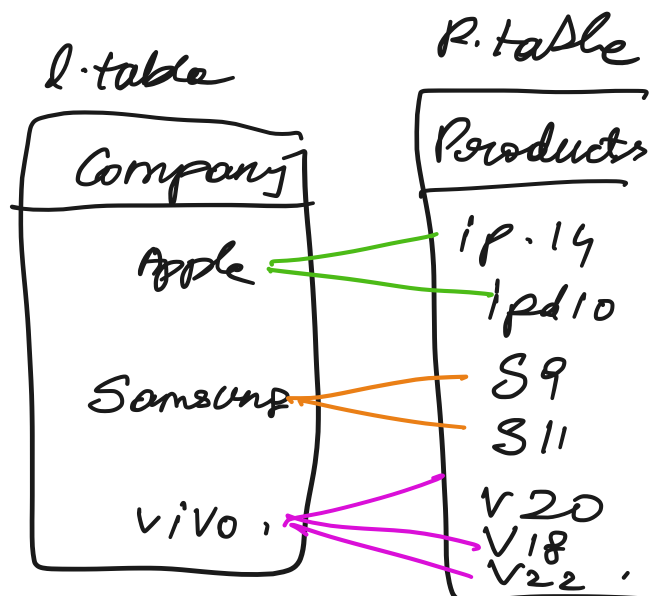
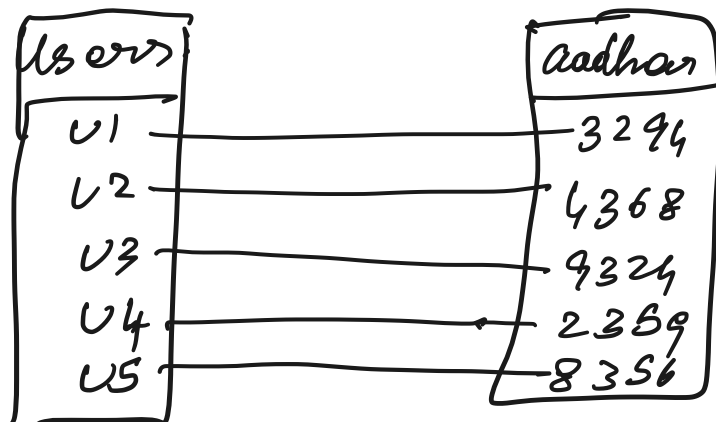
many - 1

many - many.

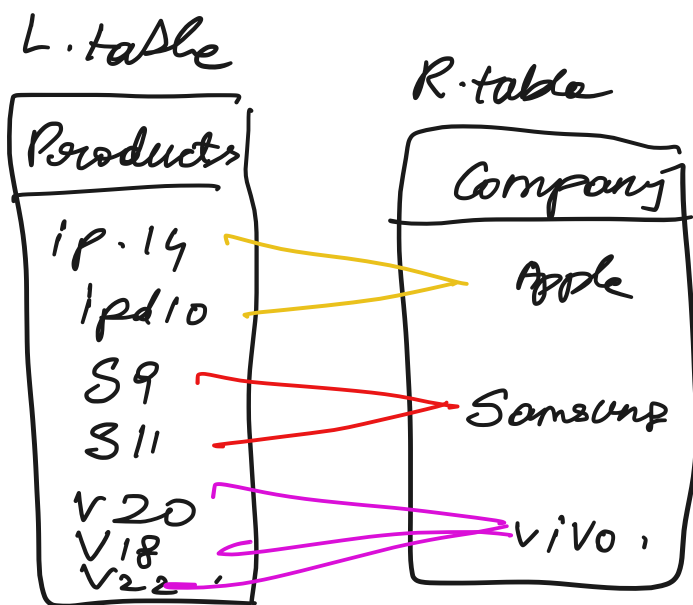
1 - 1

1 - many.

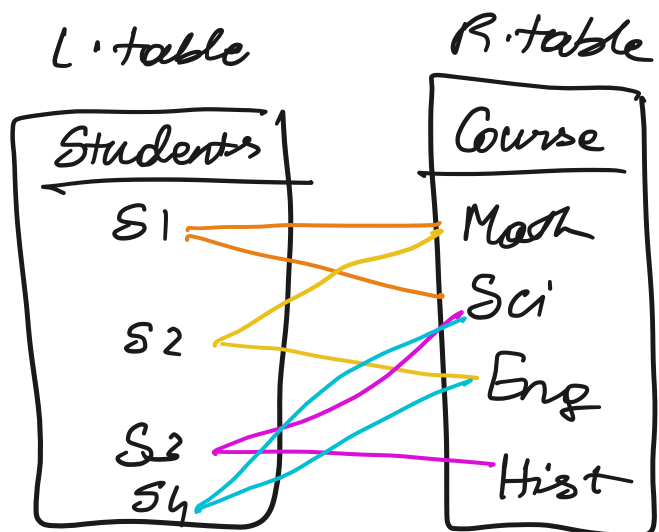
L. Table Right table



Many to 1.



Many to many.



Symbol one ——— ||
many ——— <



milky bar
panner
curd

dairy

Fname table		Lname table	
Cid	Fname	Cid	Lname
1	Thanish	1	Batch
2	John	2	Alan
3	Sowio	3	Alex

instead can be a single table

Cid	Fname	Lname
1	Thanish	Batcha
2	John	Alan
3	Sowio	Alex

Why no m-m?

Author	name	Books	bookname
A1	ABC	B1	book1
A2	XYZ	B2	book2
		B3	book3

Name → 1 MB
5 MB.

book → 2 MB

	name		bookname
A1	ABC	B1	book1
A1	ABC	B2	book2
A1	ABC	B3	book3

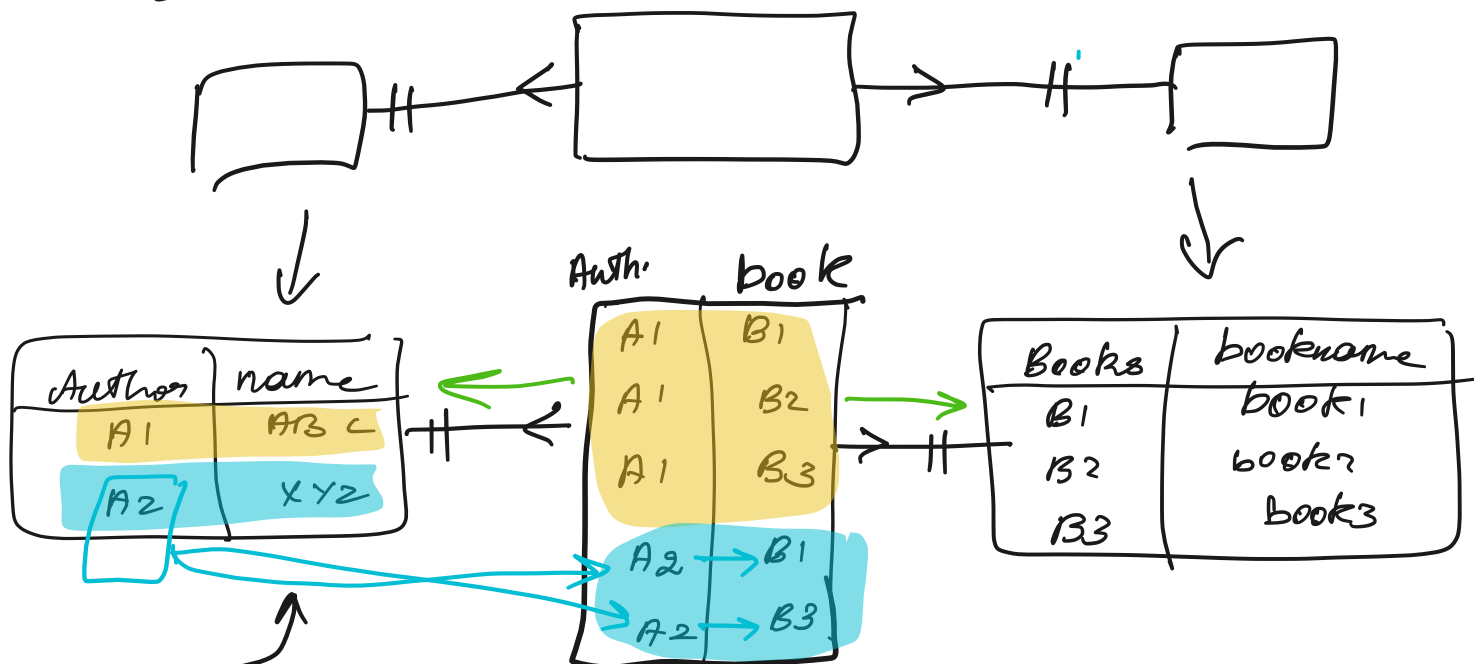
56KBs → 10MB

A2	XYZ	B1	book1
A2	XYZ	B3	book2

L-table

Junction table

R-table



Find books written by author XYZ.

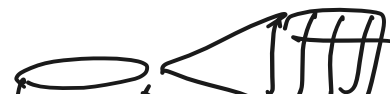
Get all rows and columns.

Syntax

Select *

from DB.table

→ give me all rows and columns from the table in the DB.



select * from
 farmer-market - product



Select * from DB-table

give all rows & columns in a table.

Ts.id	Date	P.id	Cus.id	...	...	Qty.	Amount
⋮							

billions of rows.

CEO.

Date, Sales

Select date, Sales
 from DB-table

Brand manager

Date, Sales, Brand.

Select date, brand Sales
 from DB-table.

P.name	Price.
Milk	10
meat	10

Sort by
 P.name

P.name	Price
egg.	2
meat	10

egg	2.
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milk	10
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Select * from
DB. table
Order by P-name

→ give me all data
sorted in the ascending
order of P-name

Order by P-name desc

default: Ascending.

P-name	Price.
Milk	10
meat	10
egg	2.

order by Price desc,
P-name

order by Price desc,
P-name desc

Pname	Price
meat	10
milk	10
egg	2

Pname	Price
milk	10
meat	10
egg	2

Limit: Top few rows are queried.

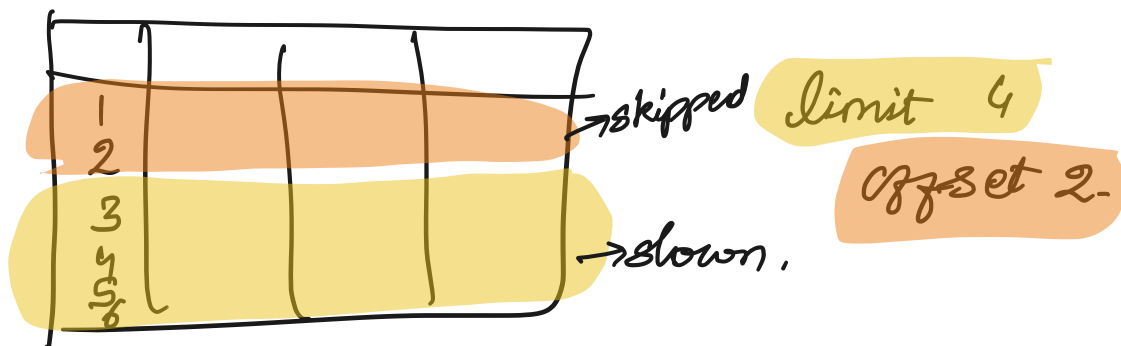
Select * from
DB. table
limit N

→ Select all the
columns but only
the top N rows
from the table.

Offset: Skip the rows and show the no. of rows
by the limit.

Select * from
DB. table
limit 2 offset 1

→ Skip 1 row
and show
2 rows.

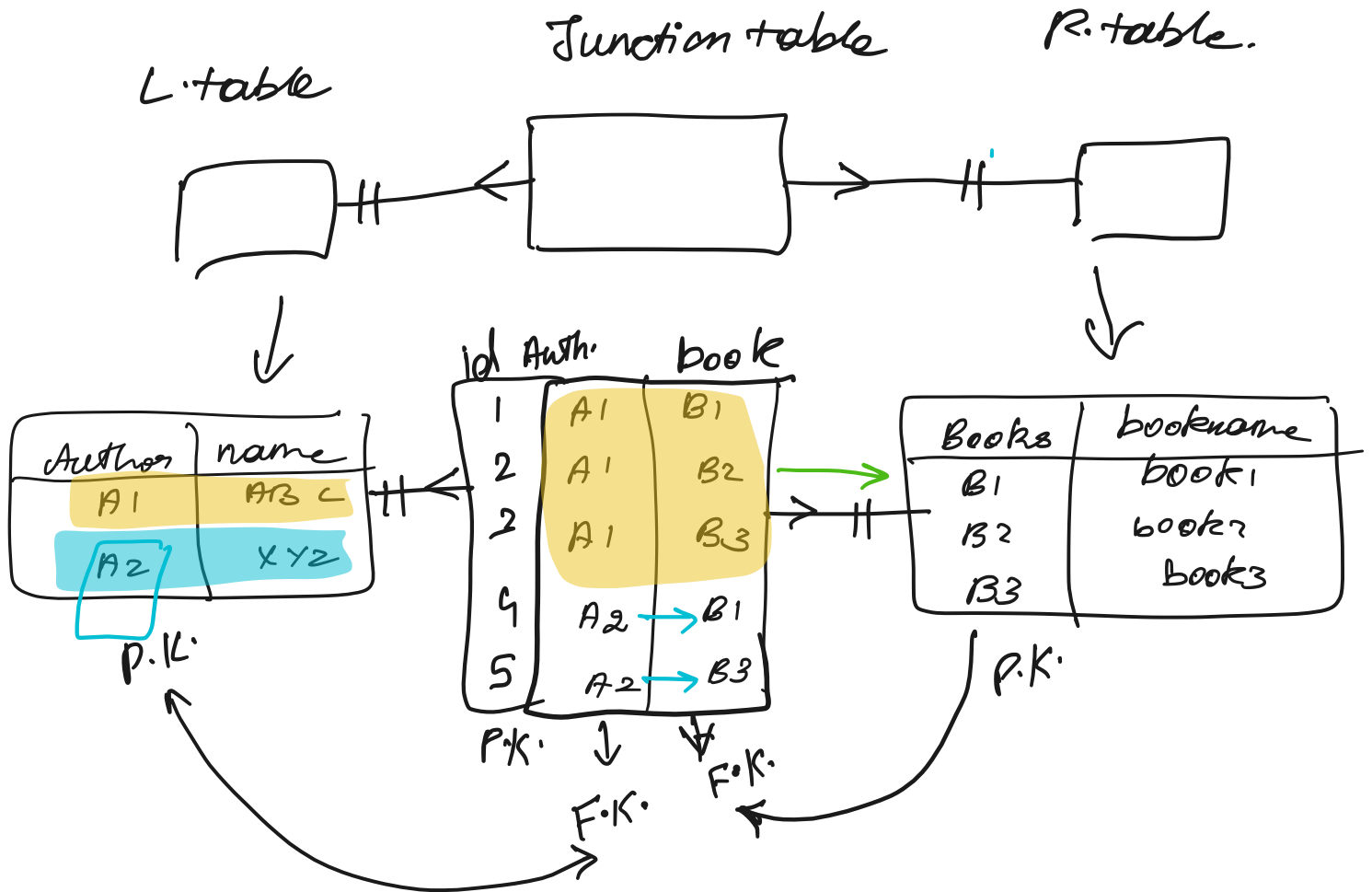


Order of writing

order of execution.

Select
From
Order by
limit
offset

From ←
Select
Order by
offset ←
limit.



Tra.id	Date	time	Amt
1	Jan 10	5:00	10
2	Jan 1	10:00	40
		1:00	50

Tra.id	date	time	Amt
3	Jan 20	1:00	50
4	Jan 11	10:00	60
1	Jan 10	5:00	10

3	Jan 20	1:00	40
4	Jan 11	10:00	60

1	Jan 1	10:00	40
2	Jan 1	10:00	40